Class 10

Physics

Volume – 2

Solutions

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2. ELECTRO MAGNETISM

WORKSHEET – 1

LEVEL – I

1. D	2. D	3. A	4. D	5. NO	6.A	7.C
8. C	9. C	10. C	11. C	12. D		

LEVEL – II

1. A	2. A	3. B	4. A	5. C	6. B	7. B
8. C	9. D	10. A	11. A	12. A	13. D	14.A
15. B	16. B	17. B	18. C	19. B		

LEVEL – III

20. A, B, C	21.A	22. D	23. A	24. C	25. a-q,r ; b-p,r: c-p,r; d-q
26. A, B, C, D		27. B	28. A	29. A	30. D	31. A
32. A	33. a-p; b-s; c-r;d-q		34. D	35. D	36. a-q; b-r; c-s; d-p

LEVEL – I

9. Given $q_1 = q_2$

$V_1 = V_2$; $B_1 = B_2$

$R = \frac{mv}{Bq}$, but $K \cdot E = \frac{1}{2}mv^2$

$$v^2 = \frac{2\ K \cdot E}{m} = \frac{2ev}{m}$$

$$R = \frac{\sqrt{m^2v^2}}{Bq} = \frac{\sqrt{m^2\frac{2ev}{m}}}{Bq} = \frac{\sqrt{2mev}}{Bq}$$

$$R\alpha\sqrt{m} \Rightarrow \frac{R_1^2}{R_2^2}$$

10. $F\alpha\sqrt{V}; \frac{F_1}{F_2} = \sqrt{\frac{v}{2v}}; F_2 = \sqrt{2}\ F$

11. $\frac{1}{2}mv^2 = qv$

$\therefore r = \sqrt{\frac{2qv}{m}}$ and $F = qvB$

12. $F = qvB\sin\theta$

LEVEL – II

1. $\bar{F} = q(\bar{V} \times \bar{B})$

$$\bar{F} = -4 \times 10^{-6} \begin{vmatrix} i & j & k \\ 2 & -3 & 1 \\ 2 & 5 & -3 \end{vmatrix} \times 10^{6} \times 10^{-2}$$

$$\bar{F} = -4 \times 10^{-6}[i(4) - j(-8) + k(16)] \times 10^{6} \times 10^{-2}$$

$$\bar{F} = -4[4i + 8j + 16k] \times 10^{6} \times 10^{-2}$$

$$\bar{F} = (-0.16\hat{\imath} - 0.32\hat{\jmath} - 0.64\hat{k})N$$

2. $F = qvB\sin 60^\circ = 4.65 \times 10^{-15}$

$$= 1.6 \times 10^{-19} \times v \times 3.5 \times 10^{-3} \times \frac{\sqrt{3}}{2}$$

$$4.6 \times 10^{-15} = 1.6 \times 10^{-22} \times 3.5 \times v \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow v = \frac{4.6 \times 10^{-15} \times 10^{22} \times 2}{1.6 \times 3.5 \times \sqrt{3}} = \frac{9.2k \times 10^{7}}{5.6 \times 1.732}$$

$$V = 0.947 \times 10^{7} = 9.47 \times 10^{6}\ m/s$$

3. $F = BqV(\theta = 90^\circ)$

$$F = 0.8 \times 2 \times 1.6 \times 10^{-19} \times 10^{5}$$

$$F = 2.56 \times 10^{-14}\ N$$

4. $\bar{F} = q(\bar{V} \times \bar{B})$

$$\bar{F} = -1.6 \times 10^{-19} \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ 3 & -15 & 0 \end{vmatrix} \times 10^{+6} \times 10^{-2}$$

$$\bar{F} = -1.6 \times 10^{-15}[i(0) - j(0) + k(-39)]$$

$$\bar{F} = (6.24 \times 10^{-14}\ N)\hat{k}$$

5. $F = BqV\sin\theta$

$$F = 3 \times 1.6 \times 10^{-19} \times 10^{7} \times \frac{1}{2}$$

$$F = 2.4 \times 10^{-12}\ N$$

6. $R = \frac{\sqrt{2mk}}{Bq};\ R \propto \frac{\sqrt{k}}{B}$

$$\frac{R_1}{R_2} = \sqrt{\frac{k_1}{k_2}}\frac{B_2}{B_1}; \frac{R_1}{R_2} = \sqrt{\frac{k}{2k}}\frac{3}{B}\frac{B}{}$$

$$\frac{R}{R_2} = \frac{3}{\sqrt{2}}; R_2 = \frac{\sqrt{2}R}{3}$$

7. $R = \frac{\sqrt{2mk}}{Bq}; R = \sqrt{\frac{2mV}{q}} \cdot \frac{1}{B}$

$$18 \times 10^{-2} = \sqrt{\frac{2 \times 9.1 \times 10^{-31} \times 2 \times 10^{3}}{1.6 \times 10^{-19}}} \cdot \frac{1}{B}$$

$$B = 8.38 \times 10^{-4}\ T$$

8. $V = \frac{E}{B};\ \frac{Bqr}{m} = \frac{E}{B}$

$$\frac{(1) \times 20 \times 10^{-3} \times 5 \times 10^{-2}}{20 \times 10^{-6}} = \frac{E}{(1)}$$

$E = 50\ V/m$

9. $x = V\sin\theta \cdot T; x = v\sin\theta \cdot \frac{2\pi m}{Bq}$

$\therefore\ \left[\frac{2\pi mv\sin\theta}{Bq}, 0, 0\right]$

10. $r = \frac{\sqrt{2mk}}{Bq}$ (k and B are constants)

$m_p = m; m_{duet} = 2\ m; m_\alpha = 4\ m$

$q_p = e; q_{duet} = e; q_\alpha = 2e$

$r\alpha \frac{\sqrt{m}}{q}; r_{p\alpha} \frac{\sqrt{m}}{e}$

$r_{duet} \propto \frac{\sqrt{2\ m}}{e}; r_\alpha \propto \frac{\sqrt{4\ m}}{2e} \propto \frac{2\ m}{2e}$

$\therefore r_\alpha = r_p < r_d$

11. $\bar{F} = q(\bar{V} \times \bar{B})$

$$\bar{F} = -1.6 \times 10^{-19} \begin{vmatrix} i & j & k \\ 2 & 2 & 0 \\ 0 & 5 & 0 \end{vmatrix} \times 10^6$$

$\bar{F} = -1.6 \times 10^{-13}[k(10)]$

$\bar{F} = -1.6 \times 10^{-13}\hat{k}$

$|\bar{F}| = +1.6 \times 10^{-12}\ N$

12. $\bar{F} = q(\bar{V} \times \bar{B})$

$$-1.28 \times 10^{-13}\hat{k} = 1.6 \times 10^{-19} \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ B_x & B_y & B_z \end{vmatrix} \times 10^6$$

$-1.28 \times 10^{-13\hat{\imath}} = 1.610^{-13}\left[i(3\ B_z) - 2\ B_z\hat{\jmath} + k\left(2\ B_y - 3\ B_x\right)\right]$

Compare $\hat{k}$ components both sides

$2\ B_y - 3\ B_z = -0.8 \Rightarrow 3\ B_x - 2\ B_y = 0.8$

$\therefore$ magneticfield is along - ve y direction

$B_y = -0.4\hat{\jmath}$

13. We know that $\bar{F} \perp \bar{B}$

$\therefore\ \bar{F} \cdot \bar{B} = 0$

$m\bar{a} \cdot \bar{B} = 0; \bar{a} \cdot \bar{B} = 0$

$[x\hat{\imath} + 2\hat{\jmath}] \cdot [3\hat{\imath} + 4\hat{\jmath}] \times 10^{-2} = 0$

$3x + 8 = 0;\ x = \frac{-8}{3}\ m/s^2$

14. $\bar{F} = q(\bar{V} \times \bar{B})$

$$\Rightarrow \bar{F} = 2 \times 10^{-6} \begin{vmatrix} i & j & k \\ 2 & 0 & 0 \\ 0 & 2 & 4 \end{vmatrix} \times 10^{6} \times 10^{-1}$$

$$\bar{F} = 0.2[i(0) - \hat{j}(8) + k(4)]$$

$$\bar{F} = [0.8\hat{k} - 1.6\hat{j}]\text{N}$$

15. $R = v\cos\theta \frac{2\pi m}{Bq}$

$$R = \sqrt{\frac{2k\theta}{m}} \cos\theta \frac{2\pi m}{Bq}$$

$$R = \frac{\sqrt{km}}{Bq} \cos\theta(2\pi m)$$

$$0.1 = \frac{\sqrt{2 \times 2 \times 10^{3} \times 10^{-19} \times 9.1 \times 10^{-31}}}{B \times 1.6 \times 10^{-19}} \times \frac{1}{2} \times 2 \times 3.14$$

$$B = 4.73 \times 10^{-3} \text{ T}$$

16. $F = BqV = (1) \times 2 \times 1.6 \times 10^{-19} \times 3 \times 10^{7}$

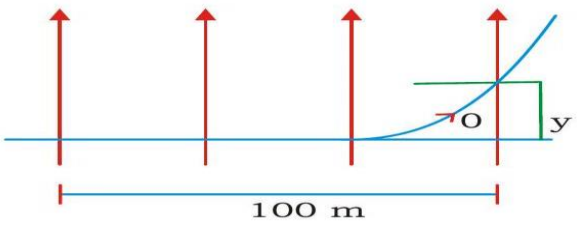
$F = 9.6 \times 10^{-12}$ N towards west

17. $a = \frac{Bq^2}{m}; y = \frac{1}{2}at^2; y = \frac{1}{2}\frac{Bq^2}{m}t^2$

$100 = vt \Rightarrow 100 = 270t \Rightarrow t = 0.37$

$$y = \frac{1}{2}\frac{500 \times 10^{-6} \times 4 \times 10^{-6} \times 270}{10 \times 10^{-3}} t^2$$

$y = 3.7 \times 10^{-6}$ m



18. $1^{H^1}, 1^{H^2}, 2^{He}; r = \frac{mv}{Bq} = \frac{\sqrt{2\ km}}{Bq}$

$$\frac{r_p}{r_d} = \sqrt{\frac{m_p}{m_d}} \times \frac{q_p}{q_d} = \sqrt{\frac{m}{2\ m}} \frac{1}{1} = \frac{1}{\sqrt{2}}; \frac{r_p}{r_\alpha} = 1$$

19. $v_p = 4 \times 10^{5}$ m/s and $m_p = 1.67 \times 10^{-27}$ kg

$B = 0.3$ T; $\theta = 60°$

$v_{perpendicular} = v\sin\theta = 4 \times 10^{5} \times \sin 60°$

$$v_{parallel} = v\cos\theta = 4 \times 10^{5} \times \cos 60^{0} = 4 \times 10^{5} \times \frac{1}{2} = 2 \times 10^{5} \text{ m/s} = 4 \times 10^{5} \times \frac{\sqrt{3}}{2}$$

$$= 2\sqrt{3} \times 10^{5} \text{ m/s}$$

$r_{perpendicular} = 1.2$ cm;

$$T = \frac{2\pi r}{V} = \frac{2 \times 3.14 \times 1.2}{2\sqrt{3} \times 10^5}$$

$$\text{pitch} = v_{\text{parallel}} \times T = 2 \times 10^5 \times \left(\frac{2 \times 3.14 \times 0.012}{2\sqrt{3} \times 10^5}\right) = 4.4 \text{ cm}$$

LEVEL – III

20. $F = mB$

$$B = \frac{F}{m} = \frac{N}{A - m}$$

$$B = \frac{Q}{A} = \frac{Wb}{m^2}$$

Options- A, B, C

21. $F = qvB$

$$F \propto v$$

$$B = \frac{N}{A - m}$$

$$[B] = \frac{[MLT^{-2}]}{[AL]} = [MT^{-2}A^{-1}]$$

22,23,24 $\bar{F} = q(\bar{V} \times \bar{B})$

$$(4\hat{\imath} + 3\hat{\jmath}) \times 10^{-16} = \begin{vmatrix} i & j & k \\ v_x & v_y & v_z \\ 0 & 0 & 4 \end{vmatrix} \times 10^{-4}$$

$(4\hat{\imath} + 3\hat{\jmath}) \times 10^{-16} = 10 \times 10^{-13}[i(4y) - \hat{\jmath}(4x)]$ compare $\hat{\imath}$ and $\hat{\jmath}$ components both sides

$4y \times 10^{-2} = 4; y = 100$

$-4x = 300; x = -75$

$\therefore \bar{v} = [-75\hat{\imath} + 100\hat{\jmath}]\text{m/s}$

27. $F = q(V \times B); F = qvB\sin\theta$

$F = qvB(\because \theta = 90°)$

$F = (4 \times 10^{-6})(270)(500 \times 60^{-6}) = 2 \times 27 \times 10^{-8}$

$F = 54 \times 10^{-8} N$

28. $a = \frac{F}{m} = \frac{54 \times 10^{-8}}{10 \times 10^{-3}}$

$a = 54 \times 10^{-6} \text{ m/s}^2$

29. $r = \frac{mv}{qB} = \frac{10 \times 10^{-3} \times 270}{4 \times 10^{-6} \times 500 \times 10^{-6}} = \frac{27 \times 10^{-1}}{2 \times 10^{-9}}$

$r = 13.5 \times 10^8 \text{ m}, d = 100 \text{ m}$

$d < r, \text{Sin}\,\theta = \frac{d}{r} = \frac{1000}{13.5 \times 10^8}$

$\text{Sin}\,\theta = 7.4 \times 10^{-8}$

$\theta = \sin^{-1}(7.8 \times 10^{-8}) \approx 0°.(0.0000044691)$

30. We know that $\bar{F} = q(\bar{V} \times \bar{B})$

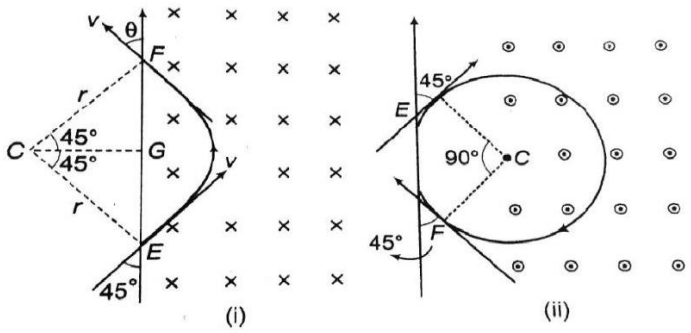
$\therefore\ \bar{F} \perp \bar{B} \Rightarrow \bar{a} \cdot \bar{B} = 0$

$(x\hat{\imath} + 4\hat{\jmath} + 2\hat{k}) \cdot (3\hat{\imath} + 2\hat{\jmath} + \hat{k}) \times 10^{-2} = 0$

$3x + 8 - 20 = 0; 3x = 12; x = 4\ \text{m/s}^2$

31. Inside a magnetic field, speed of charged particle does not change. Futher, velocity is perpendicular to magnetic field in both the cses hene path of the particle in the magnetic field will be circular. Centre of circle can be obtained by drawing perpendiculars to velocity (or tantent to the circular path) at E and F. Radius and angular speed of circular path would be

$r = \frac{mv}{Bq}$ and $\omega = \frac{Bq}{m}$ Refer figure (i)



$\angle CFG = 90° - \theta$ and $\angle CFG = 90° - 45° = 45°$ ($\because CF = CE$)

$\therefore \angle CFG = \angle CEG$ or $90° - \theta = 45°$ or $\theta = 45°$

Further $FG = GE = r\cos 45°$

$\therefore EF = 2FG = 2r\cos 45° = \dfrac{2mv\cos 45°}{Bq}$

$\Rightarrow EF = \dfrac{2(1.6 \times 10^{-27})(10^7)\left(\frac{1}{\sqrt{2}}\right)}{(1)(1.6 \times 10^{-19})} = 0.14\ \text{m} = 14\ \text{cm}$

32. Refer figure ii In this casse particle will complete $\frac{3^{\text{th}}}{4}$ of circle in the magnetic field.

Hence, the time spend in the magnetic field

$t = \frac{3}{4}$ (time period of circular motion)

$t = \dfrac{3}{4}\left(\dfrac{2\pi m}{Bq}\right) = \dfrac{3\pi m}{2q}$

$t = \frac{(3\pi)(1.6 \times 10^{-27})}{(2)(1)(1.6 \times 10^{-19})} = 4.7 \times 10^{-8}\text{sec}$

34. $R = 5\text{ cm}; r = \frac{mv}{Bq}$

$5 \times 10^{-2} = \frac{9.1 \times 10^{-31} \times 1.41 \times 10^{6}}{B \times 1.6 \times 10^{-19}}$

$B = 1.6 \times 10^{-4}\text{ T}$

35. $T = \frac{1}{2}\frac{2\pi m}{Bq} = \frac{1}{2}\frac{2\times3.14\times9.1\times10^{-31}}{1.6\times10^{-4}\times1.6\times10^{-19}}$

$T = 1.11 \times 10^{-7}\text{ s}$

WORKSHEET – 2

LEVEL – I

1. C	2. D	3. C	4. A	5. D	6. C	7. A
8. A	9. C	10. B				

LEVEL – II

1. B	2. D	3. A	4. C	5. D	6. D	7. A
8. D	9. D	10. A	11. A	12. A	13.C	14.D
15. C	16. C	17.C	18. A	19. B	20.D	21.C

LEVEL – III

22. A	23.D	24.A	25. C	26. a-2; b-1; c-5; d-3	27.A,B,C
28. D	29.B, C	30. a-q; b-r; c-p; d-s			

LEVEL – I

1. Apply the Ampere's Right hand thumb rule

2. $B = \frac{\mu_0 i}{4\pi d}[\sin\alpha + \sin\beta]$

$\Rightarrow B \propto \frac{1}{d} \Rightarrow B \propto \frac{1}{d}$, if $d = x$

3. C

4. $B = \frac{\mu_0 i}{4\pi d}[\sin\alpha + \sin\beta]$

Here ' B ' is independent of length of the conductor.

5. $B = \frac{\mu_0 i}{4\pi d}[\sin\alpha + \sin\beta]$

If $d = \infty$, then B = 0 from the axis f the solenoid.

If i = 0, Then B = 0 inside the pipe.

LEVEL – II

1. $B = \frac{M_0 i}{2\pi d} = \frac{4\pi\times10^{-7}\times i}{2\pi\times2\times10^{-2}}$

$12 \times 10^{-7} = \frac{4\pi\times10^{-7}\times i}{2\pi\times2\times10^{-2}};\ i = 1200$ A 2. $B = \frac{M_0 i}{4\pi d}[\sin 0°]; B = \frac{M_0(1)}{4\pi d}(0); B = 0$

3. $B\alpha \frac{1}{d}; \frac{B_1}{B_2} = \frac{d_2}{d_1}$

$\frac{10^{-3}}{B_1} = \frac{12}{4} = 3;\ B_2 = 0.333 \times 10^{-3}$

$B_1 = 3.33 \times 10^{-4}$ T

4. $B_H = B;$

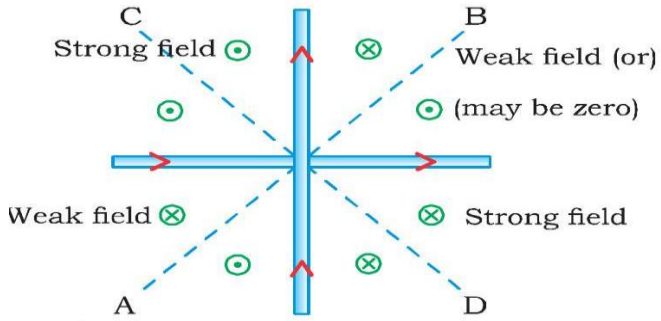
$3.6 \times 10^{-5} = \frac{M_0(3.6)}{2\pi(d)}$

$3.6 \times 10^{-5} = \frac{2 \times 3.6 \times 10^{-7}}{d}$

5. D

6. $B = \frac{\mu_0 i}{2\pi d}$

The strength of magnetic field around the conduction does not depends on diameter hence remains same.



(may be zero)

From figure along AB magnetic field is in opposite direction Along AB

8. $B = B_1 - B_2; B = \frac{\mu_0 n_1 i_1}{2r_1} - \frac{\mu_0 n_2 i_2}{2r_2}$

$B = \frac{\mu_0 n}{2}\left[\frac{i_1}{r_1} - \frac{i_2}{r_2}\right] = \frac{\mu_0(10)}{2}\left[\frac{0.2}{2 \times 10^{-1}} - \frac{0.3}{4 \times 10^{-1}}\right]$

$B = 5\mu_0\left(1 - \frac{3}{4}\right) = \frac{5\mu_0}{4}$ 9. Magnetic field due to current at centre of circles zero.

∴ Force experienced by charge is zero.

10. $B = \frac{\mu_0 i}{2\pi r} = 2 \times 10^{-7} \times \frac{10}{1} \times 10$

$B = 2 \times 10^{-5} wb/m^2$

$B = Bqv = 2 \times 10^{-5} \times 5 \times 10^6 \times 1.6 \times 10^{-19}$

$B = 1.6 \times 10^{-17} N$

11. Radius of orbit(r) $= 0.5A^0 = 0.5 \times 10^{-10}$ m

Frequency of revolution $f = 10^{10}$MHz $= 10^{16}$ Hz

Charge on electron $e = 1.6 \times 10^{-19}$C

The electron revolving in a circular orbit around the nucleus behaves as a current loop and hence posses a magnetic momentum.

The magnetic momentum can be given as M = IA

Where, I is a current flowing through the circular loop. i.e., I = e/T

T is a time period = 1/f

Therefore 1/T = f; I = ef

And A is a area of circular loop i.e., $A = \pi r^2$

$M = ef\pi r^2$

$M = 1.6 \times 10^{-19} \times 10^{16} \times 3.14 \times (0.5 \times 10^{-10})^2$

$M = 1.256 \times 10^{-23}\text{Am}^2$

12. $B = \frac{\mu_0 i}{2r} = \frac{\mu_0}{2r}\frac{e}{T} = \frac{4\times\pi\times10^{-7}\times1.6\times10^{-19}}{2\times0.5\times10^{-10}\times0.16\times10^{-15}} = 4\pi$

$B = 4 \times 3.14 = 12.56T$

13. $r = 0.4A^0 = 0.4 \times 10^{-10}$ m

$v = 10^6\ \text{ms}^{-1}, \mu_0 = 4\pi \times 10^{-7}\text{H/m}$

$e = 1.6 \times 10^{-19}\text{C}$

$v = \frac{2\pi r}{t} \Rightarrow 10^6 = \frac{2\pi \times 0.4 \times 10^{-10}}{t}$

$\Rightarrow t = 0.8\pi \times 10^{-16}$ s

$B = \frac{\mu_0 i}{2r} = \frac{\mu_0}{2r} \times \frac{e}{t} = \frac{4\pi\times10^{-7}\times1.6\times10^{-19}}{2\times0.4\times10^{-10}\times8\times\pi\times10^{-17}} = 10T$ 14. Magnetic force is given by, $F = IBl\sin\theta$

Magnetic field is perpendicular to the length of wire, so $\theta = 90°$

Magnetic force on side QR, $F_1 IBl\sin 90° = 0.2 \times 1 \times 1 = 1$ N

Magnetic force on side PR, $F_2 IBl\sin 90° = 5 \times 2 \times 0.1 = 1N$

Magnetic force on side PQ, $F_3 IBl\sin 90° = 5 \times 2 \times 0.1 = 1N$

Resultant of $\vec{F}_1$ and $\vec{F}_2$

$F' = \sqrt{F_1^2 + F_2^2 + F_1F_2\cos 120^0} = \sqrt{1+1-1} = 1N$

Resultant of $\vec{F}'$ and F_3

$F = 1 - 1 = 0$

So, net magnetic force the loop is zero.

15. The magnetic field due to an infinitely long wire at a distance d from it is given as $B = \frac{\mu_0 I}{2\pi d}$

hence the magnetic field due to the the two current carrying wires in the same direction is given as $B_{net} = B_1 - B_2$

where $B_1 = \frac{\mu_0 I_1}{2\pi d}$ where d is the distance of the midway point from either wire

while $B_2 = \frac{\mu_0 I_2}{2\pi d}$

So $B_{net} = \frac{\mu_0 I_1}{2\pi d} - \frac{\mu_0 I_2}{2\pi d}; 10 = \frac{\mu_0}{2\pi d}(I_1 - I_2)$

Similarly, the magnetic field due to the the two current carrying wires in the opposite direction is

$B_{\text{net}} = B_1 + B_2;\ B_{net} = \frac{\mu_0 I_1}{2\pi d} + \frac{\mu_0 I_2}{2\pi d}; 30 = \frac{\mu_0}{2\pi d}(I_1 + I_2)$ ...

dividing (1) by (2)

we have, $\frac{1}{3} = \frac{(I_1 - I_2)}{(I_1 + I_2)}; (I_1 + I_2) = 3(I_1 - I_2);\ \frac{I_1}{I_2} = 2$ 16. Perpendicular distance of the side of the sqaure from the centre is

$\frac{10}{2} = 5$ cm $= 0.05$ m

since direction of current is same in all sides, therefore net magnetic field will be added up

$$B = 4 \times \frac{\mu_0 I}{4\pi r}\left(\sin 45^0 - \sin(-45^0)\right)$$

$$B = 4 \times \frac{\mu_0 I}{2\sqrt{2}\pi r} = \frac{1.414 \times 4\pi \times 10^{-7}}{0.05\pi} = 113.12 \times 10^{-6} \text{ tesla}$$

17. $B = \frac{\mu_0 I}{4\pi r}\left(\frac{\pi}{2}\right)$, B due to all three arc will be mutually perpendicular

$B_{net} = \sqrt{3}\frac{\mu_0 I}{8R}$

18. $B = \frac{\mu_0 nI}{2r} = \frac{4\pi \times 10^{-7} \times 35 \times 50}{2 \times 25 \times 10^{-2}} = 1.4\pi \times 10^{-4}$

19. $B = \frac{\mu_0 nir^2}{2(r^2+x^2)^{3/2}} = \frac{1}{8} \times \frac{\mu_0 ni}{2r}$

20. $B = \frac{\mu_0 \pi n^2 i}{l} \Rightarrow B \propto n^2 \Rightarrow \frac{B_1}{B_2} = \left(\frac{n_1}{n_2}\right)^2 = \frac{16}{4} = \frac{4}{1}$

21. Magnetic field due to different parts are

$$B_1 = 0; B_2 = \frac{\mu_0}{4\pi}\frac{\pi I}{r} \odot; B_3 = \frac{\mu_0}{4\pi}\frac{I}{r} \odot$$

$$\therefore B_{\text{net}} = B_2 + B_3 = \frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r}$$

LEVEL – III

23. According to the figure the current be distributed uniformely over the crosssection of outer and inner conductors. Current density in inner conductor $\frac{i}{\pi a^2}$.

Current density in outer conductor is $\frac{i}{\pi(c^2-b^2)}$.

When r < a : Consider a coaxial circular path (Ampereon loop) of radius r. Let B be the magnitude of magnetic field at this distance, then using Ampere's law $B \times 2\pi r = \mu_0 \times$ Current enclosed by path,

$$\mathrm{B} = \mu_0\left(\frac{i}{\pi a^2} \times \pi r^2\right) = \frac{\mu_0 i r^2}{a^2}$$

$$B = \frac{\mu_0}{4\pi} \times \frac{2ir}{a^2} \text{ Wb/m}^2$$

24. When a < r < b : In this case the circular path (Ampereon loop) of radius will enclose the current passing through inner conductor. Using Ampere's law

$B \times 2\pi r = \mu_0 i$ or $B = \left(\frac{\mu_0}{4\pi} \times \frac{i}{r}\right) Wb/m^2$

25. When b < r < c : Here, current enclosed by co-axial circular path (Ampereonloop) of radius r.

Current passing through inner conductor- current passing through portions of outre conductor lying between $r = b$ and $r = c$ (-ve sign is used because the current in two conductors are in opposite direction).

By Ampere's law

$$B \times 2\pi r = \mu_0\left[i - \frac{i}{\pi(c^2-b^2)} \times \pi(r^2-b^2)\right]$$

$$B = \frac{\mu_0}{2\pi}\frac{i}{r}\left[1 - \frac{(r^2-b^2)}{(c^2-b^2)}\right] \text{Wb/m}^2 \text{ or } B = \frac{\mu_0}{2\pi}\frac{i}{x}\left[1 - \frac{(r^2-x^2)}{(c^2-b^2)}\right] \text{Wb/m}^2$$

26. One could theoretically calculate $\vec{B}_1$ define each path analytically, and perform messy integrals. Instead, use Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$

For each path, use whatever total current passes through the loop. By the righthanded-rule (fingers curl in the direction of the path, thumb points in the positive direction of positive I_{enc}), current out of the page are positive and into the page are negative. The integrals for each path are:

For path a: $\oint \vec{B}.d\vec{l} = \mu_0(0) = 0T.m$

For path b: $\oint \vec{B}.d\vec{l} = \mu_0(-I_1) = -5.15 \times 10^{-6}T.m$

For path c: $\oint \vec{B}.d\vec{l} = \mu_0(-I_1 + I_2) = 1.76 \times 10^{-6}T.m$

For path d:$\oint \vec{B} \cdot d\vec{l} = \mu_0(-I_1 + I_2 + I_3) = 4.90 \times 10^{-6} T.m$

27. The current enclosed by the first loop is $-I_1$.

The current enclosed by the second loop is $I_3 - I_2$.

The current enclosed by the third loop is $I_3 - I_2 - I_1$.

29. The field at the cntre of the loop due to the straight part is $B = \frac{\mu_0}{2\pi}\frac{I}{r}$, directed into the paper, and the field due to the loop is $B = \frac{\mu_0}{2}\frac{I}{r}$, directed out of the paper.

Hence the net field upwards $= \frac{\mu_0 i}{2R}\left(1 - \frac{1}{\pi}\right)$

WORKSHEET – 3

LEVEL – I

1. A	2. B	3. D	4. D	5. D	6. D	7. B
8. B	9. C	10.D	11. D	12. C	13.C	14. D

LEVEL – I

1.B	2.C	3.A	4.B	5.A	6.A	7.B
8.C	9.C	10.C	11.A	12.B	13.C	14.C
15.B	16.A	17.B	18.B	19.B	20.B	

LEVEL – III

21.B	22. C	23. A	24. D	25.1.6	
26. a-3, b-2, c-3, d-1	27. A, B, C, D	28. A	29. B	30. D	
31. A, B, C	32. a-4, b-5, c-2, d-1				

LEVEL – II

6. $\sin\theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$

$F = 0.1 \times 10 \times \frac{10}{100} \times \sin 53^\circ = \frac{10}{100} \times \frac{4}{5} = \frac{4}{50} = 0.08$ N

7. $F = Bil = (1) \times (2) \times 8 \times 10^{-2} [\ell = 2r] = 0.16$ N

8. $F = Bil$

$\Rightarrow 10 \times 10^{-3} \times 10^{-3} \times 8 = B \times 2 \times (1)$

$B = 4.9 \times 10^{-5}$ T

10. $\bar{F} = i(\bar{\ell} \times \bar{B})$

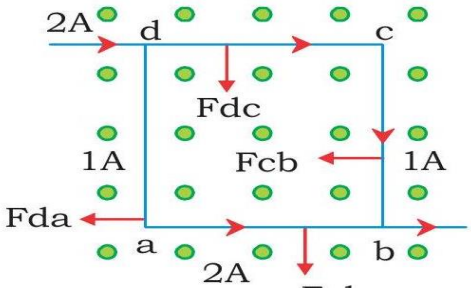
11. $F = 3Bil = 3 \times 2 \times \sqrt{2} = 36\sqrt{2}$ towards EC

12. $F = Bil \Rightarrow 0.13 = 0.067 \times i \times 0.2; i = 9.7$ A

13. $\bar{F} = i(\bar{\ell} \times \bar{B})$

$$= i\begin{vmatrix} i & j & k \\ \ell & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} B_0 = iB_0(-\hat{j} + \ell\hat{k}) = B_0 j\ell[-\hat{j} + \hat{k}] = \sqrt{2}B_0 i\ell$$

14.



Fab

$i = 2A, l = 20 \times 10^{-2}m, B = 0.1T$

$F_{ab} = Bil = 0.1 \times 20 \times 10^{-2} \times 1$

$F_{ab} = 0.02$ N down

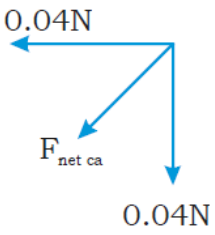
$F_{da} = 0.02$Nleft

$F_{dc} = 0.02N$ down

$F_{cb} = 0.02$ Nleft

$F_{\text{netleft}} = 0.02 + 0.02 = 0.04$ N left

$F_{\text{net down}} = 0.02 + 0.02 = 0.04$ N down



$F_{\text{netca}} = \sqrt{0.04^2 + 0.04^2}$

$= \sqrt{2}(0.04)N = 0.056$ along ca

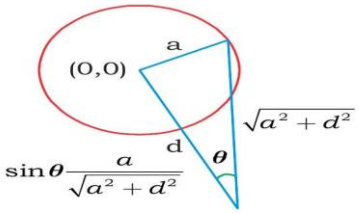
15. $\vec{F} = i(\vec{l} \times \vec{B}); \vec{F} = i\left(l\hat{\imath} \times B_0(\hat{\imath} + \hat{\jmath} + \hat{k})\right)$

$\vec{F} = ilB_0(\hat{k} - \hat{\jmath}); |\vec{F}| = ilB_0\sqrt{1^2 + 1^2} = \sqrt{2}B_0 il$ 16. $F = Bil; F = Bi(2\pi r)$

$F = Bi(2\pi a) \because (r = a)$

$F = 2\pi aiB$ into the paper

17.



$\vec{B} = B_0\vec{e}_r (\because e_r = \text{unit vector})$

$r = a; FBil\sin\theta$

$F = B_0 i(2\pi a)\dfrac{a}{\sqrt{a^2+d^2}}$

$F = \dfrac{2\pi a^2 iB}{\sqrt{a^2+d^2}}$

18. $i = neAv_d$

$v_d = \dfrac{i}{neA} = \dfrac{5}{10^{29}\times 1.6\times 10^{-19}\times 10^{-5}}$

$= \dfrac{5}{1.6}\times 10^{-5}$ m/s

$F = Bqv = \dfrac{1}{10}\times 1.6\times 10^{-19}\times\dfrac{5}{1.6}\times 10^{-5} = 5\times 10^{-25}$ N

19. Force due to electric field on charge $\vec{F}_e = \vec{E}q$

$\vec{F}_e = -102.\times 10^3\times 1.6\times 10^{-19}\hat{k}$

Force due to magnetic field on charge $\mathrm{F_m} = q(\vec{v}\times\vec{B})$

$\mathrm{F_m} = 1.6\times 10^{-19}\left((1.28\times 10^6\hat{\imath})\times(8\times 10^{-2}\hat{\jmath})\right)$

$\mathrm{F_m} = 102.4\times 10^3\times 1.6\times 10^{-19}\hat{k}$

20. $F_{mag} = q_q\left(\vec{v}\times\vec{B}_1\right)$

Magnetic induction due to charge q_1 moving with v_1 at a distance ' r ' is $B_1 = \frac{\mu_o}{4\pi}\frac{vq}{r^2}$

$F_{\text{mag}} = \dfrac{\mu_0}{4\pi}\dfrac{q_1q_2}{r^3}(\vec{v}(\vec{v}\times\vec{r}))$

$\dfrac{F_{\text{mag}}}{F_{\text{ele}}} = \dfrac{\frac{\mu_0}{4\pi}\frac{q_1q_2}{r^2}(v^2)}{\frac{\mu_0}{4\pi\varepsilon_0}\frac{q_1q_2}{r^3}} = \dfrac{1}{\mu_0\varepsilon_0}\times\dfrac{1}{v^2}$

$\dfrac{F_{\text{ele}}}{F_{\text{mag}}} = \dfrac{c^2}{v^2} \Rightarrow \dfrac{F_{\text{ele}}}{F_{\text{mag}}} = \dfrac{v^2}{c^2}$

LEVEL – III

21. The magnetic field due to wire D at wire C is

$B_D = \dfrac{\mu_0}{4\pi}\dfrac{2I}{r} = \dfrac{10^{-7}\times 2\times 30}{0.03} = 2\times 10^{-4}$ T

which is directed into the page.

The magnetic field due to wire G at C is

$B_G = \dfrac{10^{-7}\times 2\times 20}{0.02} = 2\times 10^{-4}$ T

which is directed out of the page.

Therefore, the net magnetic field at the position of the wire C is

$B = B_G - B_D = 2 \times 10^{-4} - 2 \times 10^{-4} = 0$

The force on 25 cm of wire C is

$F = Bil\sin\theta = 0$ 22,23,24.

Net field at a point will be the vector sum of the fields due to the two wires.

The magnetic field at P due to the wire a has magnitude

$B_1 = \frac{\mu_0 i}{2\pi d} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 2 \times 10^{-2}} = 1.00 \times 10^{-4}$ T

Its direction will be perpendicular to the line shown and will point downward in the figure. The field at this point due to the other wire has magnitude

$B_2 = \frac{\mu_0 i}{2\pi d} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 6 \times 10^{-2}} = 0.33 \times 10^{-4} T$

Its direction will be the same as that of B_1.Thus, the resultant field will be $1.33 \times 10^{-4} T$ also along the same direction.

Similarly, the resultant magnetic field at R will be 1.33×10^{-4} T along the direction pointing upward in the figure.

The magnetic field at point Q due to the two wires will have equal magnitudes but opposite directions and hence the resultant field will be zero.

25. magnetic field due to a straight conductor $B = \frac{\mu_0 i}{2\pi r}; B_{\text{new}} = B_1 + B_2$ (as the currents are in opposite directions)

Then $B_{\text{new}} = 2B$

$$B_{\text{new}} = 2\frac{\mu_0}{2\pi}\frac{10}{2.5 \times 10^{-2}}$$

$$= \frac{4\pi \times 10^{-5} \times 4}{\pi} = 16 \times 10^{-5} \text{ T}$$

27. Let the two wires be positioned at O&PR = OA,

$= \sqrt{(0.02)^2 + (0.02)^2} = \sqrt{8 \times 10^{-4}} = 2.828 \times 10^{-2}$ m

(a) vector B due to Q, at $A_1 = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 0.02} = 1 \times 10^{-4} T$(⊥ r towards up the line)

vector B due to P, at A1 $= \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 0.06} = 0.33 \times 10^{-4}$ T

($\perp^r$ towards down the line)

net vector $B = 1 \times 10^{-4} - 0.33 \times 10^{-4} = 0.67 \times 10^{-4}$ T (b) vector B due to O at

$A_2 = \frac{2 \times 10^{-7} \times 10}{0.01} = 2 \times 10^{-4}$ T

(⊥ r down the line)

(c) vector B at A_3 due to O $= 1 \times 10^{-4} T(\perp^r$ towards down the line)

vector B at A_3 due to P $= 1 \times 10^{-4} T(\ _\perp r$ towards down the line)

Net vector B at $A_3 = 2 \times 10^{-4}$ T

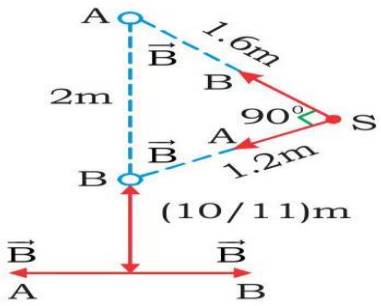
(d) vector B at A_4 due to O $= \frac{2\times10^{-7}\times10}{2.828\times10^{-2}} = 0.7 \times 10^{-4}$ T towards SE

vector B at A_4 due to P $= 0.7 \times 10^{-4} T$ towards SW

Net $\vec{B} = \sqrt{(0.7 \times 10^{-4})^2 + (0.7 \times 10^{-4})^2}$

$= 0.989 \times 10^{-4} \approx 1 \times 10^{-4} T$

31.



$i_A = 9.6$amp; $i_B = i$(say)

For P to be a neutral point, the direction of current in wire B must be just opposite to that in wire A.

(i.e., current in wire B must be in upward direction) and field produced at P due t tw current carrying wires must be of equal magnitude, i.e.,

$$\frac{\mu_0}{4\pi}\frac{2i}{AP} = \frac{\mu_0}{4\pi}\frac{i_B}{BP} \text{ (or) } \frac{2 \times 9.6}{(32/11)} = \frac{2 \times i}{(10/11)}$$

$\therefore i = 3$amp

Now, $\overrightarrow{B_A}$ = magnetic induction at S due to current $i_B = \frac{\mu_0}{4\pi} \times \frac{(2\times9.6)}{1.6}$, along $\overrightarrow{SB}$ and

$\overrightarrow{B_B}$ = magnetic induction at S due to current $i_B = \frac{\mu_0}{4\pi} \times \frac{(2\times3)}{1.2}$, along $\overrightarrow{SA}$

Because, $(AB)^2 = (SA)^2 + (SB)^2$, hence, $\angle ASB = 90°$ and therefore, net magnetic induction at S

$$|\vec{B}| = \sqrt{B_A{}^2 + B_B{}^2} = \frac{\mu_0}{2\pi}[6^2 + (2.5)^2]^{1/2}$$

$= (10^{-7} \times 2) \times 6.5 = 13 \times 10^{-7}$ T

$\frac{F}{L} =$ Force per unit length on the wire B

$$\frac{\mu_0}{4\pi} \times \frac{2i_1 i_2}{r} = \frac{10^{-7} \times 2 \times 9.6 \times 3}{2} = 2.88 \times 10^{-6}\text{Nm}^{-1}$$

WORKSHEET – 4

LEVEL – I

1. D	2. A	3. B	4. A	5. D		

LEVEL – II

1.C	2.D	3.A	4.D	5.D	6.C	7.B
8.D	9.A	11.A	12.D	13.B	14.D	15.B
16.A	17.B	18.A	19.A	20.D	21.A	22.B

LEVEL – III

23.B	24.A	25.B	26.D	27. a-1,2; b-3; c-5; d-4	28.A, B
29. a-4; b-3; c-1; d-2		30. C	31.C	32.A	

LEVEL – I

1. $\tau = I(\vec{A} \times \vec{B}) = IAB\sin 0^\circ = 0$
2. $|\mu| = NIA$
3. If the currents are in the same direction in the parallel conductors, then both will be attracts each other.
4. A
5. D

LEVEL – II

1. $\tau = NBIA = 400 \times 0.1 \times 10^{-7} \times 6 \times 10^{-4}$

   $= 2.4 \times 10^{-9}$NmC
2. $\tau = NBIA\sin\theta, \theta = 90^\circ - 60^\circ$

   $= 500 \times 10^{-3} \times 100 \times 10^{-3} \times 2 \times 10^{-4} \times \sin 30^0$

   $= 5 \times 10^{-6}$Nm
3. $\tau = BINA\sin 0^\circ = 0$
4. $\tau = BINA\ \sin 0^\circ = 0$
5. $\tau = 1 \times 0.01 \times 10 \times 0.1 = 0.01$Nm 6. $2\pi r = L \Rightarrow r = \frac{L}{2\pi}$

   $A = \pi r^2 = \frac{\pi \times L^2}{4\pi^2} = \frac{L^2}{4\pi}$ $\therefore \mu = IA = \frac{iL^2}{4\pi}$
7. $\tau = 100 \times 9 \times 10^{-4} \times 0.1 \times 0.1 = 9 \times 10^{-4}$Nm

   $\therefore$ couple per unit twist $= \frac{9\times10^{-4}}{10} = 9 \times 10^{-5}$Nm/deg ree
8. $B = \frac{\tau}{NIA} = \frac{0.2}{100\times2\times20\times10^{-4}} = 0.5$ T
9. $\tau = Bil \times d_{perpendicular}$

   $= 0.2 \times 5 \times 10 \times 10^{-2} \times 20 \times 10^{-2} = 0.02$Nm

   and force is zero
10. $i = \frac{q}{t} = \frac{q}{\left(\frac{2\pi}{\omega}\right)} = \frac{q\omega}{2\pi}$

$$\mu = \frac{iA}{2} = \frac{q\omega}{2\pi} \times \frac{\pi R^2}{r} = \frac{qR^2\omega}{4}$$

11. $t = \frac{2\pi r}{v} = \frac{2\pi \times 5.3 \times 10^{-11}}{2.2 \times 10^6}$

$$= \frac{20 \times 5.3}{7} = 1.5 \times 10^{-16}\ s$$

12. $i = \frac{1.6 \times 10^{-19}}{1.5 \times 10^{-16}} = 1.066 \times 10^{-3}\ A = 1.066\ mA$

13. $\mu = NIA = 1.1 \times 10^{-3} \times \frac{22}{7}(5.3 \times 10^{-11})^2$

$$= 9.7 \times 10^{-24} Am^2$$

14. $\vec{\tau} = \vec{\mu} \times \vec{B} = i(\vec{A} \times \vec{B})$

$$= 0.2 \times \pi \times 64 \times 10^{-4}[(0.6\hat{i} - 0.8\hat{j}) \times (0.25\hat{i} + 0.3\hat{k})]$$
$$= 0.2\pi \times 64 \times 10^{-4}(-0.18\hat{j} + 0.2\hat{k} - 0.24)$$
$$= (-9.6\hat{i} - 7.2\hat{j} + 8\hat{k}) \times 10^{-4} Nm$$

15. $w = \mu B(-\cos\theta_1 - \cos\theta_2)$

$$0.8 = 0.02 \times 52(-\cos\theta_1 - \cos\pi)$$
$$\Rightarrow -\cos\theta_1 + 1 = \frac{0.8}{0.02 \times 52} = 0.77$$
$$\Rightarrow -\cos\theta_1 = 1 - 0.77 = 0.23 \Rightarrow \theta_1 = 76.7^0$$

16. $i = \frac{ev}{2\pi r}, \mu = \frac{ev}{2\pi r} \times \pi r^2 = \frac{evr}{2}$

$$\frac{\mu}{L} = \frac{-evr}{2mvr} = \frac{-e}{2\ m}$$

17. $T_{max} = 50 \times 5 \times \pi \times 1 \times 10^{-4} \times 0.2$

$$6.3 \times 10^{-2} \sin\theta = \frac{6.3 \times 10^{-2}}{2}$$

$$\sin\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

18. $T = 500 \times 1 \times 3.14 \times 4 \times 10^{-4} \times \sin 30^0 \times 0.4$

$$= 5 \times 3.14 \times 4 \times 0.4 \times \frac{10^{-2}}{2};$$

$$\approx 0.13 Nm$$

19. $F = nbi\ell \frac{Mg \times \ell}{2} = nBi\ell \times \ell; B = \frac{Mg}{2ni\ell}$

20. $\tau = NIAB\cos\theta$

21. $|F| = \frac{\mu I_1 I_2}{2\pi} \ell \left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

$$= 2 \times 10^{-7} \times 2 \times 5 \times 10^{-1}\left(\frac{1}{10^{-2}} - \frac{1}{5 \times 10^{-2}}\right)$$
$$= \frac{2 \times 10^{-8}}{10^{-2}}\left(1 - \frac{1}{5}\right)$$
$$= 1.6 \times 10^{-5}\ N$$

LEVEL – III

24. The magnetic moment is given by: $M = nIA$

    Here n = 1 turn; I = 4 A; $A = \pi r^2 = \pi \times (0.5)^2 m^2$

    $M = 1 \times 4 \times \pi \times (0.5)^2 = 3.14\text{Am}^2$

25. If thw normal to the plane of the coil makes an angle of 90° to the direction of field, then torque (τ) is

    $\tau = MB\cos 90^\circ = 3.14 \times 2 \times 1 = 6.28\text{Nm}$

26. $P \cdot E = -M \cdot B = -MB\text{COS}\,90^\circ = 0$

27. The magnitude of the magnetic moment is given by:

    $M = nIA$

    Here n = 25 turns; I = 1.5 A;

    $A = \pi r^2 = \pi \times (5.5 \times 10^{-2})^2$

    $M = (25) \times (1.5) \times \pi \times (5.5 \times 10^{-2})^2 = 0.36\text{Am}^2$

    In vector notation, torque is given by:

    $\vec{\tau} = \vec{M} \times \vec{B}$

    The maximum torque occurs when $\vec{M}$ is perpendicular to $\vec{B}$

    $\vec{\tau}_{\max} = MB\sin 90^0 = (0.36) \times (5 \times 10^{-5}) \times 1$
    $= 1.8 \times 10^{-5}\text{Nm}$

3. ELECTRO MAGNETIC INDUCTION

WORKSHEET – 1

LEVEL – I

1. D	2. D	3. A	4. D	5. D	6. B	7. B
8. A						

LEVEL – II

1.C	2. B	3. A	4. B	5. B	6. A	7. B
8. B	9. D	10. C	11. D	12. C	13. C	14. A
15. B	16. A	17. C	18. D	19. B	20. A	

LEVEL – III

21. A, B, C, D	22. D	23. B	24. A	25. B	
26. a-r; b- p; c- q; d-s,t	27. 5	28. A, B, C	29. B	30. A, B	31. A
32. D					

LEVEL – II

1. $\phi = \text{B.A}\ \ \phi' = \text{BAcos}\,180$

$\phi = \text{BA}\ \phi' = -\text{BA}$

$e = -\dfrac{d\phi}{dt}\ \phi' = -\phi$

$\text{e} = -\dfrac{(-\phi - \phi)}{\text{t}} = \dfrac{2\phi}{\text{t}}$

$\text{i} = \dfrac{\text{e}}{\lambda} = \dfrac{2\phi}{\text{tR}}$

$\dfrac{\text{q}}{\text{t}} = \dfrac{\text{q}}{\text{tR}}\ \text{q} = \dfrac{2\phi}{\text{R}}$

2. $\text{e} = -\dfrac{\text{dd}}{\text{dt}}\ \phi = \text{B.A}$

$= \text{BAcos}\,0$

$\phi\text{i} = \text{NBA}$

$\phi\text{t} = \text{NBAcos}\,180$

$= -\text{NBA}$

$e = -\dfrac{(\text{dt} - \phi\text{i})}{\text{t}} = -\dfrac{(-\text{NBA} - \text{NBa0})}{\text{t}}$

$$e = \frac{2NBA}{t}$$

$$i = \frac{e}{R} = \frac{2NBA}{Rt}$$

$$i = 2 \times \frac{1200 \times 4 \times 10^{-4} \times 500 \times 10^{-4}}{20 \times 0.1}$$

$$i = 2400000 \times 10^{8} = 0.024a = 24\ mA$$

3. $B = \mu_0 H = \frac{\mu_0 \times 5 \times 10^4}{\pi}$

$$e = \frac{NBA}{time} = \frac{50 \times 2 \times 10^{-2} \times 10^{-2}}{0.1} = 0.11$$

4. By using $e = -\frac{NBa(\cos\theta_2 - \cos\theta_1)}{\Delta t}$

$$e = -\frac{1000 \times 2 \times 10^{-5} \times 500 \times 10^{-4}(\cos 180° - \cos 0°)}{0.2} = 10^{-2}\ volt = 10mV$$

5. The current induced will be

$$i = \frac{|e|}{R} \Rightarrow i = \left(\frac{1}{R}\right)\frac{d\phi}{dt}\ But i\ = \frac{dq}{dt}$$
$$\Rightarrow \frac{dq}{dt} = \frac{1}{R}\frac{d\phi}{dt} \Rightarrow \int dq = \frac{1}{R}\int (d\phi) \Rightarrow q = \frac{BA}{R}$$

6. Let the velocity of each point due to rotation from an angle θ with the vertical EMF induced across $PQ = \int_0^{\pi/2} B(r\omega\cos\theta)(rd\theta)$

$$= \int_0^{\pi/2} Br^2\omega\cos\theta d\theta$$
$$= B\omega r^2$$

7. Suppose r is the radius of the coil and B is the magnetic field.

Now the area of the coil is

$$A = \pi r^2$$

Magnetic flux associated is

$$\phi = nAB = n\pi r^2\ B$$

Length of the wire is $1 = 2\pi rn$

So if we make coil of radius of $2r$ then number of turns n ' will be

$$2\pi \times 2r \times n' = 2\pi rn$$

$$n' = \frac{n}{2}$$

Area of the new coil will be

$A' = \pi(2r)^2 = 4\pi r^2$

So flux associated with it will be

$\phi' = \mathrm{n'A'B} = \frac{\mathrm{n}}{2} \times 4\pi \mathrm{r}^2 \times \mathrm{B} = 2\phi$

8. $\because\ i = \frac{e}{R}$

$= \frac{\mathrm{N}(\Delta\phi/\Delta\mathrm{t})}{\mathrm{R}} = \frac{\mathrm{NS}(\Delta\mathrm{B}/\Delta\mathrm{t})}{\mathrm{R}} = \frac{10(10 \times 10^{-4})(10^4)}{20}$

$= 5\ \mathrm{A}$

9. The induced emf in the coil is $\mathrm{e} = -\mathrm{N}\left(\frac{\mathrm{do}}{\mathrm{dt}}\right)$ given

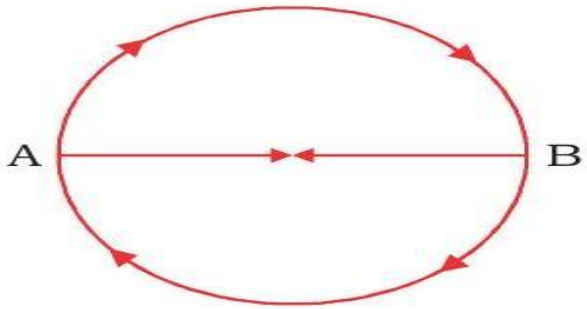
$\mathrm{dO}| = 0|_2 - 0|_1 = 2 \times 500_{\mu}\mathrm{Wbdt} = 20 \times 10^{-3}\mathrm{sec\ N} = 200$

$\mathrm{E} = -\mathrm{n}\left(\frac{\mathrm{O}|_2 - \mathrm{O}|_1}{\mathrm{dt}}\right)$ Putting the given values in Eq. (i)

$\mathrm{e} = -\frac{200 \times 2 \times 500 \times 10^{-6}}{20 \times 10^{-3}} = -10\mathrm{volt} \Rightarrow |\mathrm{e}| = 1\ \mathrm{ovolt}$

10. $\frac{\mathrm{NA}(\mathrm{B}_2 - \mathrm{B}_1)}{\mathrm{time}} = \frac{-\mathrm{NA}(-\mathrm{B} - \mathrm{B})}{\mathrm{time}} = \frac{2\mathrm{NAB}}{\mathrm{time}} \Rightarrow \frac{2 \times 200 \times 100 \times 10^{-4} \times 0.25}{0.01} = 10^2\ \mathrm{V}$

11. As per Lenz's law, emf *s* of the same magnitude are induced in the two loops in the clockwise direction. Thus current is induced in the clockwise direction in the outer boundary, there ii zero current in wire AB.



12. $\vec{\mathrm{F}}_{\mathrm{m}} \perp \vec{\mathrm{v}}$

$\therefore\ \theta = 90° + 30° = 120°$

13. The emf of the circuit is $\varepsilon = \mathrm{Blv}$

The current $I = \frac{\varepsilon}{R} = \frac{Btv}{R}$

For equilibrium of rod, $\mathrm{mg} = \mathrm{BH} = \mathrm{B} \times \frac{\mathrm{Blv}}{\mathrm{R}} \times 1$

$\mathrm{v} = \frac{\mathrm{mgR}}{\mathrm{B}^2 1^2}$

14. $\mathrm{q_A} = +80\mu\mathrm{C}$ and $\mathrm{q_B} = -80\mu\mathrm{C}$

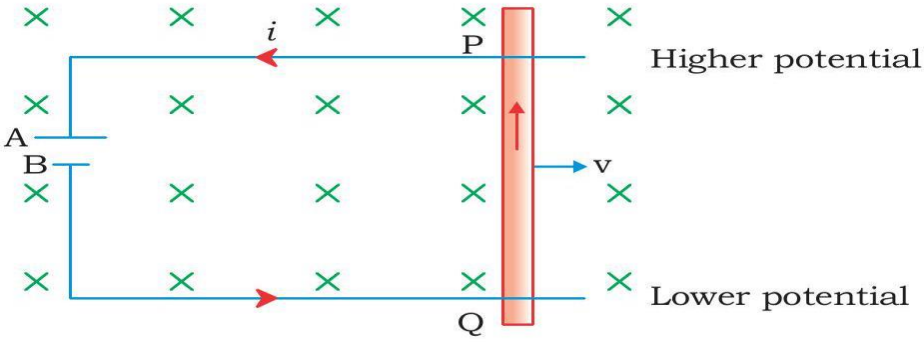
$\mathrm{q_A} = -80\mu\mathrm{C}$ and $\mathrm{q_B} = +80\mu\mathrm{C}$

$q_A = 0 = q_B$

Charge stored in the capacitor increase exponentially with time

$Q = CV{-}{=} C(Bvl) = 10 \times 10^{-6} \times 4 \times 2 \times 1 = 80\mu C$

According to Fleming's right hand rule, induced current flows from Q to P. Hence P is higher potential and Q is at lower potential therefore A is positively charge and B is negatively charged



15. $\phi = \vec{B} \cdot \vec{A}$

$= \left[(40\hat{i} - 15\hat{k})10^{-4}\right] \cdot \left[(5 \times 10^{-4}\hat{k})\right] = -750\text{nWb}$

16. $e = A\frac{dB}{dt} = 10^{-2} \times 3 \times 10^{-4} = \times 10^{-6}$

17. For $r < R, e = \int E \cdot ds, = \frac{d\phi_B}{dt}$

$E.2\pi r = -A\left(\frac{dB}{dt}\right), E.2\pi r = -\pi r^2\left(\frac{dB}{dt}\right)$

$E = -\frac{r}{2}\left(\frac{dB}{dt}\right), E = -\frac{r}{2}\alpha, (\bar{E}) = \frac{r}{2}\alpha$, Ear

18. Only vertical lines of earths magnetic field will cut when the rod is allowed to slide over the rails. Let the vertical componets of the earths magnetic field is B_v Induced current $i = \frac{B_v l v}{R}$

$3 \times 10^{-6} = \frac{B_v \times 0.3 \times 0.2}{0.2}$

$B_v = 10^{-5}$ T

given $B_H = 0.3 \times 10^{-4}$ T

The angle of dip θ is given by $= \tan^{-1}\left(\frac{B_v}{B_H}\right) = \tan^{-1}\left(\frac{1}{3}\right)$

19. Let r is radius of the loop. The magnetic flux, at a time t, through the loop is given by

$\phi = B(\pi r^2), e = \frac{d\phi}{dt} = \pi r^2\left(\frac{dB}{dt}\right)$

(the numerical value of induced emf)

induced current in the loop is given by

$$i = \frac{e}{R} = \frac{\pi r^2}{R}\frac{dB}{dt}$$

Where R is the resistance of the wire let a is the cross - section are and the length of the wire. Then

$$R = \rho\frac{1}{\pi a^2}$$

where $1 = 2\pi r, m = \pi a^2 1\sigma$

On putting the value of a^2 in eq. (ii) we get

$$i = \frac{m}{4\pi\rho\sigma}\frac{dB}{dt}$$

20. Magnetic field in the inside region of outer coil due to current 1 in the coil $= \frac{\mu_0 I}{2b}$

Thus the flux passing through the coil $= BA = \frac{\mu_0 I}{2\ b} \times \pi a^2 = \frac{\mu_0 I \pi a^2}{2\ b}$

Hence the emf produced in inner coil $= \frac{d\phi}{dt} = \frac{\mu_0 \pi a^2}{2\ b}\frac{dI}{dt}$

Hence the total charge flowing in the inner coil $= \int_0^t i_{\text{inner}}\, dt = \frac{\mu_0 \pi a^2}{2Rb}\int_0^t \frac{dI}{dt} dt$

$$= \frac{\mu_0 \pi a^2 i}{2Rb}$$

LEVEL – III

21. Length of the wire in the form of square = 40 cm

side of the square $= \frac{40}{4}$ cm = 10 cm = 0.1 m

∴ area of the loop a = 0.1 m × 0.1 m = 0.01 m^2

if B is the field at an instant the flux through the loop ϕ = ba. As the area remains constant

$$\therefore\ \varepsilon = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA) = -A\frac{dB}{dt}$$

$$= -0.01\ m^2 \times 0.02 t/s = -2 \times 10^{-4} v$$

current induced $i = -\frac{\varepsilon}{R}$

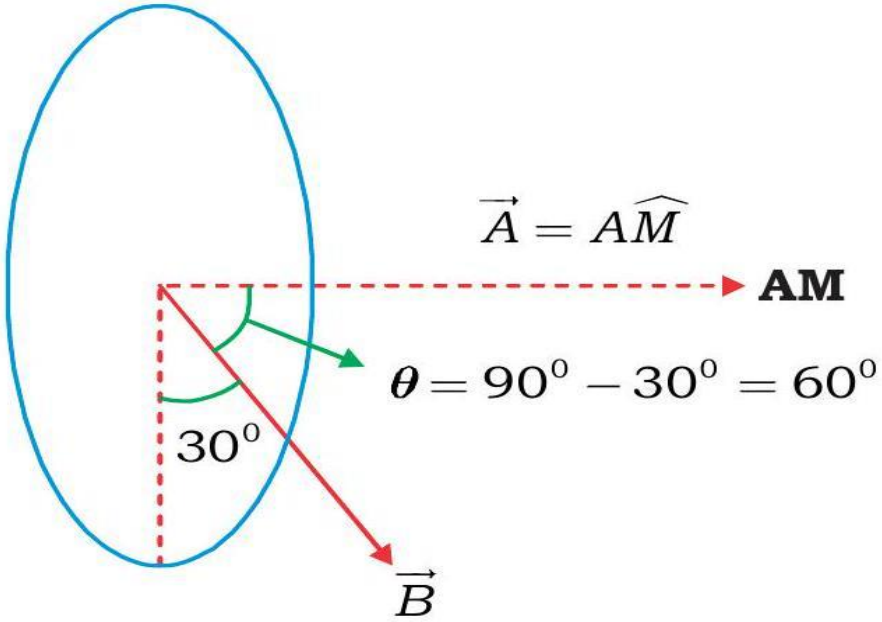
the resistance of the loop $r = \rho\frac{1}{A}$

$$\therefore\ r = \frac{1.7 \times 10^{-8}\Omega m \times 0.4\ m}{\frac{22}{7} \times (1 \times 10^{-3})^2\ m^2} = 2.16 \times 10^{-3}\Omega$$

$$\therefore\ i = -\frac{2 \times 10^{-4}}{2.16 \times 10^{-3}} a$$

$$= -9.3 \times 10^{-2} a$$

23. Body $\phi_{\text{cal}} = N(\vec{B} \cdot \vec{A}) = NBA\cos\theta$



hence N = 1, B = 10^{-4} T, A = πr^2

$$= \frac{22}{7} \times \left(\frac{21}{2} \times 10^{-2}\right)^2$$

$\theta = 90°, -30° = 60°$

$$\phi_{\text{coil}} = (1)[(10^{-4})]\left[\left(\frac{22}{3} \times 10^{-2}\right)^2\right]\cos 60°$$

$= \sqrt{3} \times 10^{-6}$ water

$= 1.732 \times 10^{-6}$ water ($\sqrt{3} = 1.732$)

24. Initial area of the loop

$A_1 = 15 \times 75$ cm^2 $= 75 \times 10^{-4}$ m^2

Initial flux density of $B_1 = 0.8$ Wb/m^2

Initial flux $(\phi_m)_1 = B_1A_1 = 0.8 \times 75 \times 10^{-4} = 0.006$ Wb

final area of loop $A_2 = 10 \times 10 = 100$ cm^2 $= 100 \times 10^{-4}$ m^2

Final flux density $B_2 = 1.4$ Wb/m^2

Final flux $(\phi_{m2}) = B_2\ A_2 = 0.014$ Wb

change in flux = $(\phi_m)_2 = (\phi_m)_2 = 0.014 - 0.006 = 0.008$ Wb

Time during which this chage in flux takes place t = 0.5 s

Induced emf = $\frac{\text{change in flux}}{\text{time}} = \frac{0.008}{0.5} = 0.016$ V $= 1.6 \times 10^{-2}$ V

25. By faradays law of emf, $e_{\underset{(average)}{cal}} = -\left[\frac{\phi_{land} - \phi_{constant}}{t}\right]$

hence, $\phi_{constant} = (NBA\cos\theta)_{initial} = (2000)(0.4 \times 10^{-4})(500 \times 10^{-4})(\cos 0°)$

$= 4 \times 10^{-3}$ water

$\phi_{land} = (NBA\cos\theta)_{land} = (200)(0.4 \times 10^{-2})(500 \times 10^{-4})\cos(180°)$

$\therefore e_{\underset{(\text{ constant })}{cal}} = -\left[\frac{(-4\times10^{-3})-(4\times10^{-2})}{\left(\frac{1}{10}\right)}\right] = 8 \times 10^{-2}$ volt $= 0.08$volt

26. a- r; b- p; c- q; d-s,t

27. Given

Induced emf, $e = 20mV = 20 \times 10^{-3}$ V

area of the loop, $A = (2 \times 10^{-2})^2 = 4 \times 10^{-4}$ m^2

Time taken to rotate the loop, $\Delta t = 0.2$ s

The average induced emf is given by

$$e = -\frac{\Delta\phi}{\Delta t} = \frac{\phi_t - \phi_1}{t}$$
$$\phi = \vec{B} \cdot \vec{A} = BA\cos\theta$$
$$\phi_1 = B(4 \times 10^{-4})\cos 0 = B(4 \times 10^{-4})$$
$$\phi_1 = B(4 \times 10^{-4})\cos 180° = -B(4 \times 10^{-4})$$
$$e = \frac{B(4 \times 10^{-4}) - |-B(4 \times 10^{4})|}{0.2}$$
$$20 \times 10^{-2} = \frac{8\ B \times 10^{-4}}{2 \times 10^{-1}}$$
$$\Rightarrow 20 \times 10^{-3} = 4 \times B \times 10^{-2}$$
$$\Rightarrow B = \frac{20 \times 10^{-3}}{4 \times 10^{-3}} = 5\ T$$

28. No of turns = 50, area of square = 0.25 m^2

Magnitude of magnetic field = 1 T

(1) average rate of complete removal $= \Delta s/\Delta t = 1$ m/s^2

$\therefore$ average induced emf $= NB\frac{S}{t} = 50 \times 1 \times 1 = 50$ volt

(2) average rate of complete restoration $= \Delta s/\Delta t = 1$ m/s^2

$\therefore$ average induced emg $= NB\frac{S}{t} = 50 \times 1 \times 1 = 50$ volt

(3) For entire motion in the field $\Delta S = 0$

$\therefore$ average induced emf $= 50 \times 1 \times 0 = 0$ volt

29. here $A = 20 \times 30\ cm^2$

$= 600 \times 10^{-4}\ m^2 = 6 \times 10^{-2}\ m^2$

$B = 0.3\ T$

$\theta_1 = s(90° - 30°) = 60°, \theta_2 = 90° - 0° = 90°$
$\phi_1 = BA\cos\theta_1 = 0.3 \times 6 \times 10^{-2}\cos 60°$

$= 0.9 \times 10^{-2}\ Wb$

$\phi_2 = BA\cos\theta_2 = 0.3 \times 6 \times 10^{-2}\cos 90° = Zero$

30. (1) Consider a coil with radius of $r = 4.00$ cm, and containing $N = 500$ turns. it is placed perpendicular in a magnetic field of B, which is given by.

$B = (0.0120\ T/s)t + (3.00 \times 10^{-5}\ T/s^4)t^4$

the magnitude of the induced emf is, $|\varepsilon| = \frac{Nd\phi_B}{dt}$

where $\phi_B = BA\cos(\phi) = BA$. Since the area in constant, then

$|\varepsilon| = NA\frac{d}{dt}(B)$

$= NA\frac{d}{dt}[(0.0120\ T/s)t + (3.00 \times 10^{-5}\ T/s^4)t^4]$

$= Na[(0.0120\ T/s) + (1.20 \times 10^{-4}\ T/s^4)t^3]$

$= (500)\pi(0.040)^2[(0.0120\ T/s) + (1.20 \times 10^{-4}\ T/s^4)t^3]$

$= 0.03.2\ V + (3.02 \times 10^{-4}\ V/s^3)t^3$

$|\varepsilon| = 0.03.2\ V + (3.02 \times 10^{-4}\ V/s^3)t^3$

(2) the induced emf at $t = 5.00s$, is

$|\varepsilon| = 0.0302\ V + (3.02 \times 10^{-4}\ V/s^3)(5.00\ s)^3$

$= 0.0680\ V$

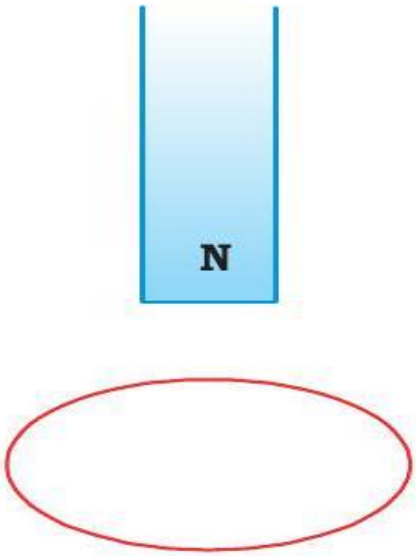
If the resistance eof the coil is $R = 600\Omega$, then the induced current at $t = 5.00$ s, is

$I = \frac{\varepsilon}{R} = \frac{0.0682\ V}{}600\Omega = 1.13 \times 10^{-4}\ A$

$I = 1.13 \times 10^{-4}\ A$

31. D

32. The north pole of a magnet is brought clown along the axis of a horizontal circular coil (figure 38.E4). As a result, the flux through the coil changes from 0.35 weber to 0.85 weber in an interval of half a second. Find the average emf induced during this period. Is the induced clockwise or anticlockwise as you look into the coil from the side of the magnet?



$\phi_1 = 0.35$ weber, $\phi_2 = 0.85$ weber

$D\phi = \phi_2 - \phi_1 = (0.85 - 0.35)$ weber $= 0.5$ weber

$dt = 0.5$sec

$E = \frac{d\phi}{dt'} = \frac{0.5}{0.5} = 1v$

The induced current in anticlockwise as seen from above.

WORKSHEET – 2

LEVEL – I

1. A	2. A	3. B	4. B	5. A	6. A	7. D
8. C	9. A	10. C	11. A			

LEVEL – II

1.A	2. C	3. A	4. A	5. D	6. B	7. C
8. A	9. D	10. D	11. B	12. B	13. B	14. D
15. B	16. B	17. B	18. D	19. C		

LEVEL – III

20. a- s; b- q; c- p; d—r 21. a- q,s; b- p,r; c- p,r; d-q,s 22. i-1; ii-1; iii-3

23. .4C 24. A, B, C, D 25. C, D 26. A, C, D

LEVEL – II

1. Induced e.m.f $\varepsilon = \frac{d\phi}{dt} = \frac{dBA}{dt} = A_0 \frac{dB}{dt}$

$= A_0 \left(\frac{4B_0 - B_0}{t}\right) = 3 A_0 B_0/t$

2. $\phi = 8t^2 + 3t + 5$

$\therefore$ we know that, induced emf, $e = \frac{d\phi}{dt}$

$$e = \frac{d(8t^2 + 3t + 5)}{dt}$$

$e = 16t + 3$

at $t == 4sec$, $e = 16 \times 4 + 3$

$e = 64 + 3$; $e = 67$ units

3. Here, $A = 10^{-2}$ m^2, $B == 10^3$ Wb/m^2

$\theta = 0°, \phi = ?$

$\phi = BA\cos\theta = 10^3 \times 10^{-2}\cos 0° = 10$ Weber

4. $i = \frac{e}{R}$

$$= \frac{N(\Delta\phi/\Delta t)}{R} = \frac{NS(\Delta B/\Delta t)}{R}$$

$= \frac{10(10\times10^{-4})(10^4)}{20}$ {1 gauss /sec $= 10^{-4}$Tesla/sec}

$= 5$ A

5. Here, $N = 500, a = 10 \times 10$ cm^2 $= 10^2 \times 10^{-4}$ m^2 $= 10^{-2}$ m^2

$\frac{dB}{dt} = 1.0$ T/s, e =?

$e = N\frac{d\phi}{dt} = NA\frac{dB}{dt} = 500 \times 10^{-2} \times 1.0 = 5$volt

6. Emf induced across the rod AB is

$e = \vec{B} \cdot (\vec{l} \times \vec{v})$

$Blv\sin\theta = 2 \times 2 \times 2 \times \sin 30$

$e = 4$ V

7. R_1 and R_2 are in parallel the this combination is in series with R

Thus, Resistance $= \frac{R_1R_2}{R_1+R_2} + R$

EMF$(e) = Blv$

$$i = \frac{Blv}{\frac{R_1R_2}{R_1+R_2} + R} = \frac{Blv(R_1+R_2)}{R(R_1+R_2)+R_1R_2}$$

8. Let the velocity of each point due to rotation from an angle θ with the vertical EMF induced across PQ $\int_0^{\pi/2} B(r\omega\cos\theta)(rd\theta)$

$= \int_0^{\pi/2} Br^2\omega\cos\theta d\theta$

$= B\omega r^2$

9. Induced emf between 0 and Q is

$\frac{1}{2}B\omega(2r)^2 = B\omega 2r^2 = 2Bvr$

But net induced emf in the ring will be zero. Hence no current flows.

10. Assuming the AK and BD are infinitely long

By Faraday's law

$E = \frac{d\phi}{dt} = BvL$

$E_{HE} = BvL$

The analogous diagram is shown in the figure below resistance and capacitor are in parallel

$E - V_c = 0$

$V_C = Bvl$

and

$Q = CV = CE_{HE} = \text{const}$

$i = \frac{dQ}{dt} = 0$

11. By using $e = -\frac{N(B_2-B_1)A}{t}; e = -\frac{500(0-0.1)\times100\times10^{-4}}{0.1} = 5$ V.

12. $e = -\frac{B(B_2-B_1)A\cos\theta}{\Delta t} = -\frac{50(0.35-0.10)\times\pi\left(3\times10^{-2}\right)^2\times\cos 0^\circ}{2\times10^{-3}}$

$= 17.7$ V

13. Here, $A = 2m^2$

$\frac{dB}{dt} = \frac{(4-1)Wb/m^2}{2\text{sec}} = \frac{3}{2}T/\text{s}$

$e = \frac{d\phi}{dt} = A\frac{dB}{dt} = 2x\frac{3}{2} = 3volt$

14. If current through A increase crosses (XX) linked with coil B increase anticlockwise current induces in coil B. As shown in figure both the current produces repulsive effect.

15. $B = \mu_0 nI = 4\pi\times10^{-7}\times200\times10^2\times1.5 = 3.8\times10^{-2}T$

$\phi = Ba = 3.8\times10^{-2}\times3.14\times10^{-4} = 1.2\times10^{-5}$ Wb

When the current in the solenoid is reversed, the change in magnetic flux,

$d\phi = 2 \times (1.2 \times 10^{-5}) = 2.4 \times 10^{-5}$ Wb

$\therefore$ induced emf, $e = N\left(\frac{d\phi}{dt}\right)$

$= 100 \times \left(\frac{2.4 \times 10^{-5}}{0.05}\right) = 0.048$ V

16. $|e| = B\frac{dA}{dt}$

$|e| = B\frac{(A_2 - A_1)}{(t_2 - t_1)}$

$|e| = \frac{1\left(\frac{\pi}{4} - 0\right)}{(1-0)}$

$e = \frac{\pi}{4}V$

17. B

18. D

$\varepsilon = Bw\int_{-2/2}^{2/2} x dx$

$= Bw\left[\frac{x^2}{2}\right]_{-2/2}^{2/2}$

$= Bw\left[\frac{L^2}{8} - \frac{L^2}{8}\right]$

$\varepsilon = 0$

19. Total length of the wire in the form of square = 40 cm

$\therefore$ side of the square $= \frac{40}{4}$ cm = 10 cm = 0.1 m

$\therefore$ area of the loop a = 0.1 m × 0.1 m = 0.01 m^2

if b is the field at an instant, the flux through the loop $\phi = BA$. as the area remians constant and magnetic field changes with time.

$\therefore \in = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA) = -A\frac{dB}{dt}$

$= -0.01\ m^2 \times 0.02 t/s = -2 \times 10^{-4} v$

current induced $i = -\frac{\varepsilon}{R}$

the resistance of the loop $R = \rho\frac{l}{A}$

$$\therefore\ R = \frac{1.7 \times 10^{-8}\Omega m \times 0.4m}{\frac{22}{7} \times (1 \times 10^{-3})^2 m^2} = 2.16 \times 10^{-3}\Omega$$

$$i = -\frac{2 \times 10^{-4}}{2.16 \times 10^{-3}} a$$

$$= -9.3 \times 10^{-2} a$$

LEVEL – III

22. $u = 1$ cm/, $B = 0.6T$

(1) At $t = 2$sec, distance moved $= 2 \times 1$ cm/s $= 2$ cm

$$E = \frac{d\phi}{dt} = \frac{0.6 \times (2 \times 5 - 0) \times 10^{-4}}{2} = 3 \times 10^{-4}\ \text{V}$$

(2) At $t = 10$sec distance moved $= 10 \times 1 = 10$ cm

The flux linked does not change with time

$\therefore E = 0$

(3) At $t = 22$sec distance $= 22 \times 1 = 22$ cm

The loop is moving out of the field and 2 cm outside.

$$E = \frac{d\phi}{dt} = B \times \frac{dA}{dt}$$

$$= \frac{0.6 \times (2 \times 5 \times 10^{-4})}{2} = 3 \times 10^{-4}\ \text{V}$$

23. $\text{Q} = \frac{NBA}{R}(\cos\theta_1 - \cos\theta_2)$

$$\text{Q} = \frac{500 \times 0.2 \times 0.1(\cos\theta - \cos 180)}{50} = 0.4C$$

WORKSHEET – 3

LEVEL – I

1.A	2. B	3. D	4. B	5. D

LEVEL – II

1.B	2. D	3. C	4. A	5. B	6. D	7. D
8. A	9. A	10. D	11. B	12. D	13. C	14. B
15. B	16. A	17. A	18. B	19. D	20. A	21. B

LEVEL – II

22. C	23. C	24. D	25. C	26. a-p; b-r; c-t; d-s
27.0.05	28.2H	29. A		

LEVEL – II

1. Given $e = 250 \times 10^{-6}$ V

$di = 10 - 6 = 4A$
$dt = 0.4\text{sec}$
$L = \dfrac{e}{dI/dt} = \dfrac{250 \times 10^{-6} \times 0.4}{4}$
$= 25 \times 10^{-6} H$

2. M = ?

dI = 6 A

N = 2000

B = $12 \times 10^{-4} wb$

$\phi = BA$

$\phi = (12 \times 10^{-4})A$

$N\phi = MI$

$M = \dfrac{N\phi}{I}$

$M = \dfrac{2000 \times 12 \times 10^{-4} A}{6}$

$M = 4 \times 10^{-1}$

$M = 0.4H$

3. Given: $M = 1.5$ henry

$dl = 5A$
$dt = 1\text{mil sec} = 10^{-3}\text{sec}$

$e = M \cdot \dfrac{dI}{dt}$

$e = 1.5 \times \dfrac{5}{10^{-5}} = 7.5 \times 10^3$ V

$= 7.5 \times 10^3$ V

4. A

5. B

6. We have $N = 500$

$A = 100$ cm$^2 = 10^2$ m^2

$\dfrac{dB}{dt} = 1T/s$

$\therefore e = -NA\frac{dB}{dt}\cos\theta$, here $\theta = 0°$

$\Rightarrow e = -500 \times 10^{-2} \times 1 \times 1 = -5$ V

Thus, the magnitude of emf generated is = 5 V.

7. The induced emf in the coil is $e = -N\left(\frac{d\phi}{dt}\right)$ given

$d\phi = \phi_2 - \phi_1 = 2 \times 500\mu Wbdt = 20 \times 10^{-3}\text{sec}\, N = 200$

$e = -N\left(\frac{\phi_2 - \phi_1}{dt}\right)$ Putting the given values is Eq (i)

$e = -\frac{200 \times 2 \times 500 \times 10^{-6}}{20 \times 10^{-3}} = -10$ volt $\Rightarrow |e| = 1$ ovolt

8. Magnetic flux, $\phi = B \cdot A$

Change in flux, $d\phi = B.dA$

$= 0.05(101 - 100) \times 10^{-4}$
$= 5 \times 10^{-6}$ Wb

Charge $dQ = \frac{d\phi}{R}$
$= \frac{5 \times 10^{-6}}{2} = 2.5 \times 10^{-6}$C

9. $\varepsilon = \frac{-(\phi_2 - \phi_1)}{t} = \frac{-(0.NBA)}{t} = \frac{NBA}{t}$

$t = \frac{NBA}{\varepsilon} = \frac{50 \times 2 \times 10^{-2} \times 10^{-2}}{0.1} = 0.1s$

10. According to faraday's law voltage is induced due to change in th magnetic flux through a closed loop.

Here magnetic flux = $\phi = BA$ = const

hence,

induced EMF = $\frac{d\phi}{dt} = 0$

induced current = $i = E/R = 0$

11. The area covered by an angle θ

$A = \pi r^2 \frac{\theta}{2\pi} = \frac{r^2}{2}\theta$

The induced EMF,

$|E| = \frac{d\phi}{dt} = \frac{d(BA)}{dt}$
$= B\frac{r^2}{2}\frac{d\phi}{dt} = B\frac{r^2}{2}w$

$v = 120\text{rev/min} = 2\text{rev/s}$

$w = 2\pi v = 4\pi\text{rad/s}$

$r = 0.5m$

$B = 004G = 0.4 \times 10^{-4}T$

Therefore,

Induced emf is given by,

$$E = \frac{1}{2}Br^2w$$

$$E = \frac{2\pi \times 2 \times 0.4 \times 10^{-4} \times (0.5)^2}{2} = 0.628 \times 10^{-5}\ \text{V}$$

12. Perpendicular distance between BC and centre O is 10 cm component of induced electric filed along

The rod $= \frac{d}{2}\frac{dB}{dt}$

where d = perpendicular distance from centre to the rod.

hence, potential difference between the ends of rod

$$v = EI = i \cdot \frac{d}{2}\frac{dB}{dt}$$
$$= \frac{10}{2} \times 10^{-2} \times 20 \times 10^{-2} \times 2 = 20\text{mV}$$

13. Perpendicular C to D form O is 20 cm

$$V = \frac{d}{2} \times I \times \frac{dB}{dt} = \frac{20}{2} \times 10^{-2} \times 20 \times 10^{-2} \times 2 = 40\text{mV}$$

14. Magnetic field produced by a current I in a larger square loop at its centre.

$$B\alpha\frac{I}{L} \Rightarrow B = K\frac{I}{L}$$

∴ magnetic flux linked with the smaller loop.

$$\phi = BS \Rightarrow \phi = \left(K\frac{I}{L}\right)(l^2)$$

Therefore, the mutual inductance $M = \frac{\phi}{I} = K\frac{l^2}{L}$ or $M\alpha\frac{l^2}{L}$

15. $E = N\frac{d\phi}{dt}$

$= \frac{d}{dt}(N\phi) - - - - - (1)$

$E = L\frac{dI}{dt} = \frac{d}{dt}(LI)$----------(2

$N\phi = LI$

$\frac{500 \times 5 \times 10^{-4}}{10} = L \times 5; \text{L} = 5\text{mH}$

16. $L = \frac{\mu_0 N^2 \pi r^2}{l}$ length of wire $= N2\pi r =$ constant (=C, suppose)

$\therefore L = \mu_0 \left(\frac{C}{2\pi r}\right)^2 \frac{\pi r^2}{l}$

$\therefore L\alpha \frac{1}{l}$

∴ Self - inductance will becomes 2 L

17. A

18. Go to the rotating frame with an instantaneous angular velocity $\vec{\omega}(t)$ in this frame, a force $2m\vec{v} \times \vec{\omega}(t)$

acts which must be balanced by the magnetic force $e\vec{v} \times \vec{B}(t)$

Thus $\vec{\omega}(t) = -\frac{q}{2m}\vec{B}(t)$

19. $e = -L\frac{dI}{dt}$ $L = \frac{e}{\frac{dI}{dt}} = \frac{96 \times 10^{-3}}{1.2} = 80 \times 10^{-3}\text{H} = 80\text{mH}$

20. $M = \frac{\mu_0 \pi r^2}{2R}$

21. $\varphi_0 = N_2 B_1 A$

or $\varphi_2 = N_2 \frac{\mu_0 N_1 N_2 A}{L}$

or $\kappa_2 = \frac{\mu_0 N_1 N_2 A}{L} I - 2$

comparing with $\varphi_2 = MI_1$, weget $\frac{\mu_0 N_1 N_2 A}{L}$

LEVEL – III

22. B

23. $L_1 + L_2 + L_3 = 9H$

24. $L_1 + \frac{L_2 \times L_3}{L_2 + L_3} = 3 + \frac{3 \times 3}{3+3} = 3 + 1.5 = 4.5H$

25. $L = 5H$

$$|E| = 50 \text{ V} = L\frac{dI}{dt}$$
$$\Rightarrow dt = \frac{L \times dI}{50} = \frac{5 \times 2}{5} = 0.2\text{sec}$$

26. a-p; b-r;c-t;d-s

27. $\text{dI} = 1 - 0 = 1; \text{dt} = 0.1\text{sec};\ L = 5 \times 10^{-3}H$

$$\varepsilon = -L \times \frac{dI}{dt} = 5 \times 10^{-3} \times \frac{1}{0.1} = 5 \times 10^{-2}V$$

29. L = 1H

$$i = \frac{V}{R} = \frac{20}{10} = 2 \text{ A}$$

$$\text{Energy} = \frac{1}{2}LI^2 = \frac{1}{2} \times 1 \times 2 = 1J$$